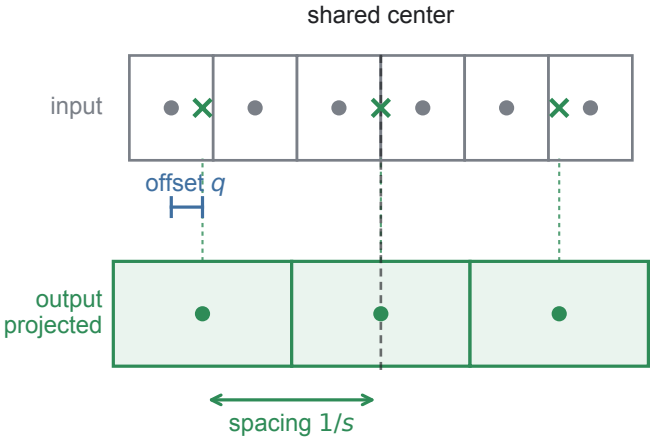
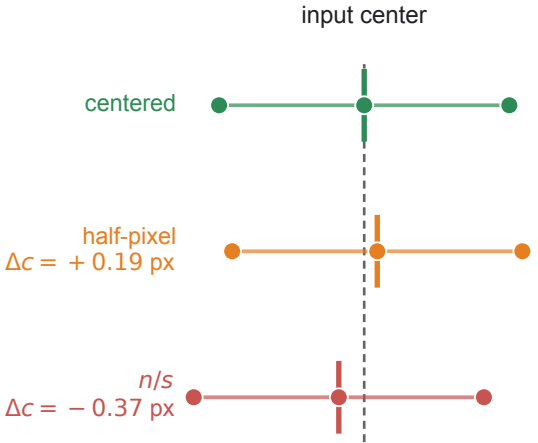
a Space the samples, then center them



$$g_n = \frac{n}{s} + q, \qquad q = \frac{N_0 - 1}{2} - \frac{N - 1}{2s}$$

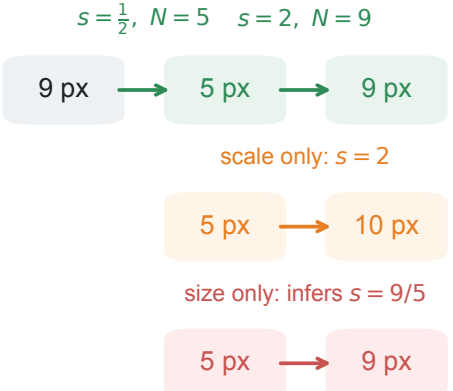
example: $N_0 = 6$, $N = 3$, $s = 0.47$

b Two shortcuts move the optical axis



half-pixel is exact only when $N = sN_0$
n/s is exact only when $q = 0$

c A round trip needs scale and size



Scale fixes spacing; size fixes extent.
Pass both to preserve the intended zoom and center.